

Transforming Quadratic Functions from Vertex Form to General Form

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A quadratic function may be written in different but equivalent forms.

Vertex Form

The **vertex form** is

$$y = a(x - h)^2 + k.$$

Basic Quadratic Form

The **basic quadratic form** is

$$y = ax^2.$$

Vertical Translation Form

The **vertical translation form** is

$$y = ax^2 + b.$$

Horizontal Translation Form

The **horizontal translation form** is

$$y = a(x - b)^2.$$

Standard Form

The **standard form** is

$$y = ax^2 + bx + c.$$

Transforming to Standard Form

To transform from **vertex form** to **standard form**, expand

$$(x - h)^2 = x^2 - 2hx + h^2,$$

multiply by a , then combine like terms.

Examples

Example 1

Rewrite

$$y = 3(x - 0)^2 + 0$$

in the form $y = ax^2$.

$$y = 3x^2$$

Example 2

Rewrite

$$y = 2(x - 0)^2 + 7$$

in the form $y = ax^2 + b$.

$$y = 2x^2 + 7$$

Example 3

Rewrite

$$y = -4(x + 5)^2$$

in the form $y = a(x - b)^2$.

$$y = -4(x - (-5))^2$$

Example 4

Rewrite

$$y = (x - 4)^2 + 6$$

in standard form.

$$y = x^2 - 8x + 16 + 6$$

$$y = x^2 - 8x + 22$$

Example 5

Rewrite

$$y = 2(x + 3)^2 - 5$$

in standard form.

$$y = 2(x^2 + 6x + 9) - 5$$

$$y = 2x^2 + 12x + 13$$

Example 6

Rewrite

$$y = -3(x - 2)^2 + 1$$

in standard form.

$$y = -3(x^2 - 4x + 4) + 1$$

$$y = -3x^2 + 12x - 11$$

Example 7

Rewrite

$$y = 5(x - 1)^2$$

in standard form.

$$y = 5(x^2 - 2x + 1)$$

$$y = 5x^2 - 10x + 5$$

Example 8

Rewrite

$$y = -2(x + 4)^2 - 3$$

in standard form.

$$y = -2(x^2 + 8x + 16) - 3$$

$$y = -2x^2 - 16x - 35$$

Example 9

Rewrite

$$y = \frac{1}{2}(x - 6)^2 + 4$$

in standard form.

$$y = \frac{1}{2}(x^2 - 12x + 36) + 4$$

$$y = \frac{1}{2}x^2 - 6x + 22$$

Example 10

Rewrite

$$y = -\frac{3}{2}(x + 2)^2 + 5$$

in standard form.

$$y = -\frac{3}{2}(x^2 + 4x + 4) + 5$$

$$y = -\frac{3}{2}x^2 - 6x - 1$$

Practice Exercises

Practice 1

Rewrite

$$y = 3(x - 0)^2 + 0$$

in the form $y = ax^2$.

Practice 2

Rewrite

$$y = 2(x - 0)^2 + 7$$

in the form $y = ax^2 + b$.

Practice 3

Rewrite

$$y = -4(x + 5)^2$$

in the form $y = a(x - b)^2$.

Practice 4

Rewrite

$$y = (x - 4)^2 + 6$$

in the form $y = ax^2 + bx + c$.

Practice 5

Rewrite

$$y = 2(x + 3)^2 - 5$$

in the form $y = ax^2 + bx + c$.

Practice 6

Rewrite

$$y = -3(x - 2)^2 + 1$$

in the form $y = ax^2 + bx + c$.

Practice 7

Rewrite

$$y = 5(x - 1)^2$$

in the form $y = ax^2 + bx + c$.

Practice 8

Rewrite

$$y = -2(x + 4)^2 - 3$$

in the form $y = ax^2 + bx + c$.

Practice 9

Rewrite

$$y = \frac{1}{2}(x - 6)^2 + 4$$

in the form $y = ax^2 + bx + c$.

Practice 10

Rewrite

$$y = -\frac{3}{2}(x + 2)^2 + 5$$

in the form $y = ax^2 + bx + c$.

Activities

Activity 1

Rewrite

$$y = -2(x - 0)^2 + 0$$

in the form $y = ax^2$.

Activity 2

Rewrite

$$y = 5(x - 0)^2 - 4$$

in the form $y = ax^2 + b$.

Activity 3

Rewrite

$$y = 3(x + 7)^2$$

in the form $y = a(x - b)^2$.

Activity 4

Rewrite

$$y = (x + 5)^2 - 8$$

in the form $y = ax^2 + bx + c$.

Activity 5

Rewrite

$$y = 4(x - 2)^2 + 3$$

in the form $y = ax^2 + bx + c$.

Activity 6

Rewrite

$$y = -2(x + 1)^2 - 6$$

in the form $y = ax^2 + bx + c$.

Activity 7

Rewrite

$$y = 6(x + 3)^2$$

in the form $y = ax^2 + bx + c$.

Activity 8

Rewrite

$$y = -5(x - 4)^2 + 2$$

in the form $y = ax^2 + bx + c$.

Activity 9

Rewrite

$$y = \frac{3}{2}(x - 2)^2 - 1$$

in the form $y = ax^2 + bx + c$.

Activity 10

Rewrite

$$y = -\frac{1}{2}(x - 8)^2 + 7$$

in the form $y = ax^2 + bx + c$.

Thank you!